

ALEXEY KHROMIN

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PROFILE

Builds production machine learning and trading infrastructure in Python and C++. Five machine learning projects spanning gradient boosting, transfer learning, computer vision and model interpretability, including a classification system processing 20,000+ monthly cash transactions in SAP. Currently building Counterback, an open-source execution-realistic limit order book backtesting engine in Python and C++, with its Binance depth-recording data pipeline already deployed on AWS and Oracle Cloud; the evaluation harness for a King's College London Finance Hub collaboration on DiffLOB, the group's generative limit order book model. Prize-winner at Citadel's Terminal algorithmic trading competition. Foundations in linear algebra, calculus, probability and statistics; automated testing throughout.

EDUCATION

King's College London

Sep 2023 – Jun 2027

MSci Computer Science · Expected First Class Honours

- **Key modules:** Compilers and Formal Languages (89%), Machine Learning (87%), Computer Systems (86%), Database Systems (84%), Foundations of Computing 2 (probability and statistics) (79%)
- **Final-year modules include:** High Frequency Finance, C++ for Financial Mathematics, Statistics in Finance, Scientific Computing for Finance, Agent Based Modelling in Finance

LSE Summer School · ME319 Machine Learning and Stochastic Simulations for Finance

Aug 2026

A-Levels · Mathematics A*, Further Mathematics A, Computer Science A, Physics B

Sep 2021 – Jun 2023

EXPERIENCE

Data Science Intern | Innoverv

Summer 2025 | London, UK

- Built and deployed a production classification system for SAP S/4HANA cash clearing, processing **20,000+ monthly cash transactions** and cutting manual matching effort by **70%+**.
- Engineered the feature pipeline from noisy transaction data: historical frequency patterns, TF-IDF text similarity between references, temporal differences and financial metadata.
- Benchmarked LightGBM against Random Forest and XGBoost on out-of-sample F1 and selected on evidence; final model reached **94% F1** and **93.6% precision**.
- Integrated SHAP so finance staff could see, per transaction, why the model matched it; historical match frequency emerged as the dominant signal.
- Separately, built a Selenium scraping pipeline with retry handling that catalogued **18,000+ SAP Fiori applications** into structured JSON for model training, replacing an estimated **4,000+ hours** of manual work.

Undergraduate Research Fellow | King's College London

Summer 2024 | London, UK

King's Undergraduate Research Fellowship · competitively awarded, funded · computer vision, embedding models

- Optimised a real-time face recognition pipeline with frame batching and multiprocessing, delivering a **30% throughput improvement** that made live operation feasible.
- Compared multi-scale embedding strategies with adaptive thresholding against fixed-threshold baselines; **97% recognition accuracy**.

PROJECTS

Counterback: Execution-Realistic Limit Order Book Backtesting Engine

Jun 2026 – present

Open source · github.com/AKhromin/Counterback

- Designing an event-driven backtesting engine with full limit order book reconstruction and price-time-priority matching; C++ matching core via `pybind11` with the strategy API, accounting and reporting in Python; three fill model fidelities from mid-spread to queue position simulation; square root market impact cost models; historical replay and reactive market modes.
- Built and deployed the market data foundation: a Binance Spot Level-2 depth recorder with two-recorder cross-verification and an offline verifier, unit tests, GitHub Actions CI on Python, Docker and systemd deployment on AWS and Oracle Cloud, and verified cloud data offload.
- Evaluation harness for a King's Informatics Finance Hub research collaboration on whether adversarial trading agents can exploit artefacts in order books generated by DiffLOB (Wang and Ventre, 2026); the engine consumes and measures generated order books rather than producing them.
- Research design published ahead of results: three pre-registered hypotheses on how the no-market-impact assumption inflates backtested performance, nine stated limitations, and a three-level validation plan including cross-checks against NautilusTrader.

Clearway: Real-Time Transport Anomaly Detection

2026

Multi-source signal fusion, anomaly detection, computer vision

- Built a real-time anomaly-detection pipeline that fuses three independent live feeds (TfL bus-arrival predictions, road-disruption reports and JamCam street imagery) and flags where official forecasts diverge from observed conditions on the Aldwych to Elephant & Castle corridor.
- Classified divergences into three mismatch types (observed but unpredicted, predicted but unobserved, unlogged congestion); scored each event on severity, confidence and timing gap, escalated it through tiered alert levels, and closed it automatically once later observations realigned with the forecast.
- YOLOv8 object detection supplies per-frame vehicle classification and traffic-density estimates; FastAPI backend with an in-memory event pipeline and route-health tracking; React, TypeScript and Leaflet front end; event logs exported to CSV and JSON for operator review rather than automated action.

Citadel Terminal Algorithmic Trading Competition

2025

- Designed and implemented a trading algorithm that won a **\$1,000 prize against 100+ competitors**.

Sign Language to Speech Translator | gesture recognition, speech models

2025

- Real-time system combining MediaPipe gesture recognition and Whisper speech models through a Flask API and a Flutter interface. Hackathon winner.

Car Make and Model Classifier | ResNet34 transfer learning, CUDA

2023

- ResNet34 transfer learning on the Stanford Cars dataset, **92% accuracy**; CUDA acceleration cut training time by **95%** versus CPU.

TECHNICAL SKILLS

- **Languages:** Python, C++, SQL, Java, TypeScript
- **Machine learning:** gradient boosting (LightGBM, XGBoost), deep learning and transfer learning (PyTorch), computer vision, model interpretability (SHAP), evaluation and benchmarking, scikit-learn
- **Data engineering:** pandas, NumPy, WebSocket APIs, Selenium, AWS, Oracle Cloud, rclone
- **Systems:** pybind11, CUDA, Linux, Git, GitHub Actions CI, Docker, systemd, unit and integration testing, Flask, React
- **Mathematics:** linear algebra, calculus, probability and statistics, hypothesis testing, Monte Carlo simulation
- **Quantitative finance:** limit order book data and reconstruction, backtesting design, execution and market-impact modelling, stylised facts, algorithmic trading

CERTIFICATIONS

- **Linear Algebra for Machine Learning and Data Science | DeepLearning.AI, 2026**
- **Calculus for Machine Learning and Data Science | DeepLearning.AI, 2026**
- **Probability & Statistics for Machine Learning & Data Science | DeepLearning.AI, 2026**
- **Machine Learning A-Z: AI, Python & R | Udemy, 2025**
- **Python for Machine Learning & Data Science Masterclass | Udemy, 2024**

AWARDS

- Citadel Terminal algorithmic trading competition, \$1,000 prize (2025)
- King's Undergraduate Research Fellowship, competitively awarded (2024)
- "Found in Translation" hackathon, winner (2025)